

SQL Assignment: Database Querying & Analysis

Instructions: Using the provided BikeStores database schema, write the SQL queries required to answer the following business and technical scenarios. Your queries should be logically sound and optimized for accuracy.

Section 1: Data Patterns and String Analysis

Scenario 1: The marketing department is running a retrospective campaign for older inventory. Provide a list of all products in the **3** or **6** categories that have the year **2016** included in their name.

Scenario 2: For data integrity auditing, identify all customers whose email addresses end with a **.com** domain but do not contain their recorded first_name anywhere within the email string.

Scenario 3: The logistics team requires a list of unique area codes for all physical stores. Extract the **first three digits** of the phone number from the sales.stores table.

Scenario 4: Identify staff members whose last names are exactly four characters long and contain the letter **'a'** as the second character.

Section 2: Conditional Logic and Range Filtering

Scenario 5: The finance department needs a list of orders that were captured in the system during the **year 2017** but were not officially marked as shipped until the **year 2018**.

Scenario 6: Locate all items in the production.products table that meet the following combined criteria:

- The list_price is between **\$500 and \$3,000**.
- The price ends specifically in **.99** (e.g., \$999.99).
- The model_year is **not** 2016.

Scenario 7: Retrieve all "Pending" or "Processing" orders (status codes **1** or **2**) that were assigned to staff members with the IDs **2, 3, or 5**.

Scenario 8: Find all customers living in the states of **New York (NY)** or **California (CA)** whose zip codes begin with the digits **10** or **90**.

Section 3: Data Aggregation and Summarization

Scenario 9: For each product category, calculate the **price gap** (the difference between the maximum and minimum list price) to help management assess pricing tiers.

Scenario 10: Generate a report showing the **total number of items** ever ordered for every unique product_id recorded in the system.

Scenario 11: Calculate the **average quantity** of stock currently held across all product categories for each individual store location.

Scenario 12: Determine which model_year saw the highest number of unique products released to the market.